UNIT-4

NON-LINEAR PROGRAMMING PROBLEM (NLPP)

Formulation of mathematical model

1) A manufacturing unit produces two products, the radio and TV sets. The production cost of each product depends upon the no. of units being processed. If x_1 and x_2 are the no. of radio and TV sets produced, Then the production costs are $200x_1 + 0.2x_1^2$ and $300x_2 + 0.2x_2^2$ respectively. There is restriction on the production capacity of the radios and TV sets to 100 and 80 units respectively. Similarly there is restriction on the man power available. A total 520 man days are available. The production of 1 piece of radio requires 2 man days and 1 TV set requires 3 man days. The sale price is dependent upon the quantity to be produced and the sale relationships are given below

Product	Quantity demanded	Unit price
Radio	$2000 - 5P$	P
TV	$4000 - 10q$	q

The problem is to determine the no. of radios and TV sets which should be produced to maximize the profit.

Soln:

Total profit is $\boxed{P = R - (C_1 + C_2)}$ where $R = Px_1 + Qx_2$

$C_1 = 200x_1 + 0.2x_1^2$, $C_2 = 300x_2 + 0.2x_2^2$

Since x_1 & x_2 are the quantities of radio and TV to be produced by the firm

$x_1 = 2000 - 5p$ | $x_2 = 4000 - 10q$

$5p = 2000 - x_1$ | $10q = 4000 - x_2$

$p = \frac{2000 - x_1}{5}$

$p = 400 - 0.2x_1$

$q = \frac{4000 - x_2}{10}$

$q = 400 - 0.1x_2$

$$R = (400 - 0.2x_1)x_1 + (400 - 0.1x_2)x_2$$

$$R = 400x_1 - 0.2x_1^2 + 400x_2 - 0.1x_2^2$$

$$\max P = 400x_1 - 0.2x_1^2 + 400x_2 - 0.1x_2^2 - 0.1x_2^2 - 200x_1 - 0.2x_1^2 - 300x_2 - 0.2x_2^2$$

$$\max P = 200x_1 + 100x_2 - 0.4x_1^2 - 0.3x_2^2$$

subject to constraints

$$2x_1 + 3x_2 \leq 520$$

$$x_1 \leq 100$$

$$x_2 \leq 80$$

$$x_1, x_2 \geq 0.$$

2) A company manufacture two products A and B. It takes 30 minutes to process 1 unit of product A and 15 minutes to process 1 unit of product B. The maximum machine time available is 35 hr/week. One unit of product A requires two kg of raw material while B requires 3 kg of raw material per unit. The available raw material is limited to 180 kg/week. The products A & B have unlimited market potential and sell for ₹200 & ₹500 per unit respectively. If the manufacturing costs of products A and B are $2x^2$ and $3y^2$ respectively. Find how much of each product should be produced per week.

$p = \frac{2000 - x_1}{5}$	$q = \frac{4000 - x_2}{10}$
$p = 400 - 0.2x_1$	$q = 400 - 0.1x_2$

$$R = (400 - 0.2x_1)x_1 + (400 - 0.1x_2)x_2$$

$$R = 400x_1 - 0.2x_1^2 + 400x_2 - 0.1x_2^2$$

$$\max P = 400x_1 - 0.2x_1^2 + 400x_2 - 0.1x_2^2 - 0.1x_2^2 - 200x_1 - 0.2x_1^2 - 300x_2 - 0.2x_2^2$$

$$\max P = 200x_1 + 100x_2 - 0.4x_1^2 - 0.3x_2^2$$

subject to constraints

$$2x_1 + 3x_2 \leq 520$$

$$x_1 \leq 100$$

$$x_2 \leq 80$$

$$x_1, x_2 \geq 0$$

2) A company manufacture two products A and B. It takes 30 minutes to process 1 unit of product A and 15 minutes to process 1 unit of product B. The maximum machine time available is 35 hr/week. One unit of product A requires two kg of raw material while B requires 3 kg of raw material per unit. The available raw material is limited to 180 kg/week. The products A & B have unlimited market potential and sell for ₹200 & ₹500 per unit respectively. If the manufacturing cost of products A and B are $2x^2$ and $3y^2$ respectively. Find how much of each product should be produced per week.

A → x
B = y

Soln:

Total revenue per week ⇒ $200x + 500y$

Manufacturing cost ⇒ $2x^2 + 3y^2$

Profit per week ⇒ $200x + 500y - 2x^2 - 3y^2$ (Total revenue − Manufacturing cost)

Subject to constraints

$$30x + 15y \leq 35 \times 60$$

$$30x + 15y \leq 2100 \qquad 2x + 5y \leq 140$$

$$2x + 3y \leq 180$$

$$x, y \geq 0$$

not in syllabus

Concave and Convex functions:

Quantity	Convex	strictly convex	concave	strictly concave
$\frac{\partial^2 f}{\partial x_1^2} \frac{\partial^2 f}{\partial x_2^2} - \left(\frac{\partial^2 f}{\partial x_1 \partial x_2}\right)^2$ (AC − B²)	≥ 0	> 0	≥ 0	> 0
$\frac{\partial^2 f}{\partial x_1^2}$	≥ 0	> 0	≤ 0	< 0
$\frac{\partial^2 f}{\partial x_2^2}$	≥ 0	> 0	< 0	< 0

Constrained optimization with equality constraints

Max (or) min $z = f(x_1, x_2)$

Subject to constraints

$$g(x_1, x_2) = b \quad ; \quad x_1, x_2 \geq 0$$

$$h(x_1, x_2) = g(x_1, x_2) - b$$

The necessary sufficient conditions

The necessary conditions for the maximum (or) minimum of $f(x_1, x_2)$, subject to the constrains $h(x_1, x_2) = 0$ can be obtained as

$$\frac{\partial L}{\partial x_1} = 0 \quad , \quad \frac{\partial L}{\partial x_2} = 0 \quad , \quad \frac{\partial L}{\partial \lambda} = 0$$

where L is the Lagrangian function can be formulated as

$$L(x_1, x_2, \lambda) = f(x_1, x_2) - \lambda h(x_1, x_2)$$

where λ is the Lagrangian multipliers

The necessary condition for optimization are given by

$$\frac{\partial L}{\partial x_1} = \frac{\partial f_0}{\partial x_1} - \lambda \frac{\partial h}{\partial x_1} = 0 \quad \text{and similarly others}$$

These necessary conditions are also the sufficient conditions for a maximum if the objective function is concave and for a minimum if the objective function is convex

1) Solve the non-linear programming problem

$$\max \; Z = 4x_1 - x_1^2 + 8x_2 - x_2^2$$

subject to constrains

$$x_1 + x_2 = 2$$

$$x_1, x_2 \geq 0$$

Soln:

Lagrangian function is given by

$$L(x_1, x_2, \lambda) = f(x_1, x_2) - \lambda h(x_1, x_2)$$

$$f(x_1, x_2) = 4x_1 - x_1^2 + 8x_2 - x_2^2$$

$$h(x) = x_1 + x_2 - 2 = 0$$

$$x_1, x_2 \geq 0$$

$$L(x_1, x_2, \lambda) = 4x_1 - x_1^2 + 8x_2 - x_2^2 - \lambda(x_1 + x_2 - 2)$$

$$\frac{\partial L}{\partial x_1} = 4 - 2x_1 - \lambda = 0 \rightarrow (1)$$

$$\frac{\partial L}{\partial x_2} = 8 - 2x_2 - \lambda = 0 \longrightarrow (2)$$

$$\frac{\partial L}{\partial \lambda} = -(x_1 + x_2 - 2) = 0$$

$$x_1 + x_2 = 2 \longrightarrow (3)$$

$(1) - (2) \Rightarrow -4 - 2x_1 + 2x_2 = 0$

$$-2x_1 + 2x_2 = 4 \longrightarrow (4)$$

$(3) \times 2 + (4)$

$$\Rightarrow 2x_2 + 2x_2 = 4$$

$$4x_2 = 8$$

$$x_2 = 2$$

$$\boxed{x_2 = 2}$$

$(3) \Rightarrow x_1 + 2 = 2$

$$x_1 = 0$$

$$\boxed{x_1 = 0}$$

$(1) \Rightarrow 4 - 0 - \lambda = 0$

$$\boxed{\lambda = 4}$$

For maximum $AC - B^2 \geq 0$ & A (or) $C < 0$.

$$\frac{\partial f}{\partial x_1} = 4 - 2x_1$$

$$A = \frac{\partial^2 f}{\partial x_1^2} = -2 < 0$$

$$B = \frac{\partial^2 f}{\partial x_1 \partial x_2} = 0.$$

$$\frac{\partial f}{\partial x_2} = 8 - 2x_2$$

$$C = \frac{\partial^2 f}{\partial x_2^2} = -2 < 0.$$

$AC - B^2 = 4 - 0$

$AC - B^2 = 4 > 0$

The Max $z = 16 - 4 = 12$.

at $x_1 = 0, x_2 = 2$.

2) Solve the NLPP Max $z = 5x_1 + x_2 - (x_1 - x_2)^2$

subject to

$x_1 + x_2 = 4$

$x_1, x_2 \geq 0$.

Soln:

$f(x_1, x_2) = 5x_1 + x_2 - (x_1 - x_2)^2$

$h(x_1, x_2) - 4 = 0$.

$L(x_1, x_2, \lambda) = 5x_1 + x_2 - (x_1 - x_2)^2 - \lambda(x_1 + x_2 - 4)$.

$\frac{\partial L}{\partial x_1} = 5 - 2(x_1 - x_2) - \lambda = 0$.

$= 5 - 2x_1 + 2x_2 - \lambda = 0 \longrightarrow (1)$

$\frac{\partial L}{\partial x_2} = 1 + 2(x_1 - x_2) - \lambda = 0$.

$= 1 + 2x_1 - 2x_2 - \lambda = 0. \longrightarrow (2)$.

$\frac{\partial L}{\partial \lambda} = -(x_1 + x_2 - 4) = 0$

$x_1 + x_2 = 4 \longrightarrow (3)$.

$(1) - (2)$

$4 - 4x_1 + 4x_2 = 0$.

$-x_1 + x_2 = -1 \longrightarrow (4)$

(3) - (4)

$2x_2 = 3$

$x_2 = 3/2$ $\quad \boxed{x_2 = 3/2}$

(3) $\Rightarrow x_1 + 3/2 = 4$

$x_1 = 4 - 3/2$

$x_1 = 5/2$ $\quad \boxed{x_1 = 5/2}$

(1) + (2) $\Rightarrow$

$6 - 2\lambda = 0$

$-2\lambda = -6$

$2\lambda = 6$

$\boxed{\lambda = 3}$

$\frac{\partial f}{\partial x_1} = 5 - 2(x_1 - x_2)$

$\frac{\partial^2 f}{\partial x_1^2} = -2 < 0$

Mam $\boxed{\frac{\partial^2 f}{\partial x_1 x_2}} = 2 = B$

$\frac{\partial f}{x_2} = 1 + 2(x_1 - x_2)$

$\frac{\partial^2 f}{\partial x_2^2} = -2 = C < 0.$

$(-2)(-2)$

$4 - 4 = 0$

$AC - B^2 = 4 - 4 = 0.$

Max $Z = 5(5/2) + 3/2 - (5/2 - 3/2)^2$

$= \frac{25}{2} + \frac{3}{2} - \left(\frac{2}{2}\right)^2$

$= \frac{25}{2} + \frac{3}{2} - 1 \Rightarrow \frac{25}{2} + \frac{3}{2} - \frac{2}{2}$

$= \frac{26}{2} = 13$ at $x_1 = 5/2$ & $x_2 = 3/2$.

3) ~~Solve the NLPP.~~

~~Min = x Max =~~

3) Solve the NLPP

Min $z = x_1^2 + x_2^2 + x_3^2$

Subject to $4x_1 + x_2^2 + 2x_3 = 14$

$x_1, x_2, x_3 \geq 0$

Soln:

$L(x_1, x_2, x_3, \lambda) = x_1^2 + x_2^2 + x_3^2 - \lambda(4x_1 + x_2^2 - 2x_3 - 14)$

$\frac{\partial L}{\partial x_1} = 2x_1 - 4\lambda = 0 \Rightarrow x_1 = 2\lambda$

$\frac{\partial L}{\partial x_2} = 2x_2 - 2\lambda x_2 = 0 \Rightarrow x_2(1 - \lambda) = 0$

$x_2 = 0$ (or) $\lambda = 1$

$\frac{\partial L}{\partial x_3} = 2x_3 - 2\lambda = 0 \Rightarrow x_3 = \lambda$

$\frac{\partial L}{\partial \lambda} = -(4x_1 + x_2^2 - 2x_3 - 14) = 0$ ——(1).

Case (1)

x_1, x_2, x_3 in (1).

$-(8\lambda + 0 - 2\lambda - 14) = 0$.

$-6\lambda = -14$

$\lambda = 7/3$

$x_1 = 14/3$
$x_2 = 0$
$x_3 = 7/3$ } (I).

case(ii)

$\lambda = 1$

$x_1 = 2, \; x_2 = 0, \; x_3 = 1 \longrightarrow$ II

$\frac{\partial f}{\partial x_1} = 2x_1$

$\frac{\partial f}{\partial x_1^2} = 2 > 0$ (A)

$\frac{\partial f}{\partial x_2} = 2x_2$

$\frac{\partial f}{\partial x_1 \partial x_2} = 0$ (B)

$\frac{\partial f}{\partial x_2} = 2x_2$

$\frac{\partial f}{\partial x_2^2} = 2 > 0$ (C)

$AC - B^2 = 4 > 0$

I $\Rightarrow$

$\min z = \frac{14^2}{9} + \frac{49}{9} = 27$

II $\Rightarrow$

$\min z = 4 + 1 = 5$ at $x_1 = 2, \; x_2 = 0, \; x_3 = 1.$

4. Determine the optimum solution for the following non-linear PP and check whether minimize (or) maximize the objective function

$Z = x_1^2 + 10x_1 + x_2^2 - 6x_2 + x_3^2 - 4x_3$

subject to constraints

$x_1 + x_2 + x_3 = 7$ and $x_1, x_2, x_3 \geq 0.$

Soln:

$f(x) = x_1^2 - 10x_1 + x_2^2 - 6x_2 + x_3^2 - 4x_3$, $h(x) = x_1 + x_2 + x_3 = 7$.

Let us determine the problem is maximization or minimization, The construct the bordered Hessian matrix as follows

$$H^B = \begin{pmatrix} 0 & P \\ P^T & Q \end{pmatrix}_{(m+n)\times(m+n)}$$

$m = 1$, $n = 3$ → 4×4

where m is the no. of constrains and n is the no of variables

where $P = \left(\frac{\partial h_1}{\partial x_1}, \frac{\partial h_1}{\partial x_2}, \frac{\partial h_1}{\partial x_3}\right)$

$P = (1, 1, 1)$ $\quad g(x) = x_1 + x_2 + x_3 - 7$

$$Q = \begin{pmatrix} \frac{\partial^2 L}{\partial x_1^2} & \frac{\partial^2 L}{\partial x_1 \partial x_2} & \frac{\partial^2 L}{\partial x_1 \partial x_3} \\ \frac{\partial^2 L}{\partial x_2 \partial x_1} & \frac{\partial^2 L}{\partial x_2^2} & \frac{\partial^2 L}{\partial x_2 \partial x_3} \\ \frac{\partial^2 L}{\partial x_3 x_1} & \frac{\partial^2 L}{\partial x_3 \partial x_2} & \frac{\partial^2 L}{\partial x_3^2} \end{pmatrix}$$

sign of H^B

Maximization $= (-1)^{m+n}$ $(-1)^4 = (1)$

Minimization $= (-1)^m = (-1)^1 = -1$

$$L = x_1^2 - 10x_1 + x_2^2 - 6x_2 + x_3^2 - 4x_3 - \lambda(x_1 + x_2 + x_3 - 7)$$

$$\frac{\partial L}{\partial x_1} = 2x_1 - 10 - \lambda, \quad \frac{\partial L}{\partial x_2} = 2x_2 - 6\lambda, \quad \frac{\partial L}{\partial x_3} = 2x_3 - 4 - \lambda$$

$$\frac{\partial^2 L}{\partial x_1^2} = 2 \qquad \frac{\partial^2 L}{\partial x_2^2} = 2 \qquad \frac{\partial^2 L}{\partial x_3^2} = 2$$

$$\frac{\partial^2 L}{\partial x_1 \partial x_2} = \frac{\partial^2 L}{\partial x_1 \partial x_3} = \frac{\partial^2 L}{\partial x_2 \partial x_1} = \frac{\partial^2 L}{\partial x_2 \partial x_3} = \frac{\partial^2 L}{\partial x_3 \partial x_1} = \frac{\partial^2 L}{\partial x_3 \partial x_2} = 0$$

$$Q = \begin{pmatrix} 2 & 0 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 2 \end{pmatrix}$$

$$H^B = \begin{pmatrix} 0 & 1 & 1 & 1 \\ 1 & 2 & 0 & 0 \\ 1 & 0 & 2 & 0 \\ 1 & 0 & 0 & 2 \end{pmatrix}$$

(0 → Q, P → Q, P^T → Q)

$$|HB| = 0 - 1\begin{vmatrix} 1 & 0 & 0 \\ 1 & 2 & 0 \\ 1 & 0 & 2 \end{vmatrix} + 1\begin{vmatrix} 1 & 2 & 0 \\ 1 & 0 & 0 \\ 1 & 0 & 2 \end{vmatrix} - 1\begin{vmatrix} 1 & 2 & 0 \\ 1 & 0 & 2 \\ 1 & 0 & 0 \end{vmatrix}$$

4 $\quad -4 \quad -4$

$= 0 - 1(4) + 1(-4) - 1(-4)$

$= -4 + 4 + 4$

$= -4$

minimization

Necessary condition

$$\frac{\partial L}{\partial x_1} = 0 \Rightarrow 2x_1 - 10 - \lambda = 0$$

$$2x_1 = 10 + \lambda$$

$$x_1 = 5 + \lambda/2 \longrightarrow (1)$$

$$\frac{\partial L}{\partial x_2} = 0 \Rightarrow 2x_2 - 6 - \lambda = 0$$

$$2x_2 = 6 + \lambda$$

$$x_2 = 3 + \lambda/2 \longrightarrow (2)$$

$$\frac{\partial L}{\partial x_3} = 0 \Rightarrow 2x_3 - 4 - \lambda = 0$$

$$x_3 = 2 + \lambda/2 \longrightarrow (3)$$

$$\frac{\partial L}{\partial \lambda} = x_1 + x_2 + x_3 - 7 = 0.$$

$$\Rightarrow x_1 + x_2 + x_3 = 7 \longrightarrow (4).$$

(1), (2), (3) in (4)

$$5 + \lambda/2 + 3 + \lambda/2 + 2 + \lambda/2 = 7.$$

$$3\lambda/2 = 7 - 10$$

$$\lambda = -3 \times 2/3$$

$$\boxed{\lambda = -2}$$

$$x_1 = 5 - 2/2 = 4.$$

$$x_2 = 2$$

$$x_3 = 1.$$

Min $Z = 16 - 40 + 4 - 12 + 1 - 4 = -35$. at $x_1 = 4 \quad x_2 = 2 \quad x_3 = 1$.

Solve the following NLPP for

min $Z = x_1^2 + x_2^2 + x_3^2$.

subject to constraints.

$$x_1 + x_2 + 3x_3 = 2.$$

$$5x_1 + 2x_2 + x_3 = 5$$

Soln:

$$f(x) = x_1^2 + x_2^2 + x_3^2 \quad \&$$

$$h_1(x) = x_1 + x_2 + 3x_3 - 2$$

$$h_2(x) = 5x_1 + 2x_2 + x_3 - 5$$

$$L(x_1, x_2, \lambda) = x_1^2 + x_2^2 + x_3^2 - \lambda_1(x_1 + x_2 + 3x_3 - 2) - \lambda_2(5x_1 + 2x_2 + x_3 - 5)$$

$$\frac{\partial L}{\partial x_1} = 2x_1 - \lambda_1 - 5\lambda_2$$

$$\frac{\partial L}{\partial x_2} = 2x_2 - \lambda_1 - 2\lambda_2$$

$$\frac{\partial L}{\partial x_3} = 2x_3 - 3\lambda_1 - \lambda_2$$

$$\frac{\partial L}{\partial \lambda_1} = -x_1 - x_2 - 3x_3 + 2$$

$$x_1 + x_2 + 3x_3 = 2 \longrightarrow (4)$$

$$\frac{\partial L}{\partial \lambda_2} = -5x_1 - 2x_2 - x_3 + 5$$

$$5x_1 + 2x_2 + x_3 = 5 \longrightarrow (5)$$

$$\frac{\partial L}{\partial x_1} = 0 \Rightarrow 2x_1 = \lambda_1 + 5\lambda_2$$

$$x_1 = \frac{\lambda_1 + 5\lambda_2}{2} \longrightarrow (1)$$

$$\frac{\partial L}{\partial x_2} = 0 \Rightarrow 2x_2 = \lambda_1 + 2\lambda_2$$

$$x_2 = \frac{\lambda_1 + 2\lambda_2}{2} \longrightarrow (2)$$

$$\frac{\partial L}{\partial x_3} = 0 \Rightarrow x_3 = \frac{3\lambda_1 + \lambda_2}{2} \longrightarrow (3)$$

Sub (1), (2), (3) in (4).

$$\frac{\lambda_1 + 5\lambda_2}{2} + \frac{\lambda_1 + 2\lambda_2}{2} + \frac{9\lambda_1 + 3\lambda_2}{2} = 2$$

$$\frac{11\lambda_1}{2} + \frac{10\lambda_2}{2} = 2$$

$$11\lambda_1 + 10\lambda_2 = 4 \longrightarrow (6)$$

(5) $\Rightarrow \frac{5\lambda_1}{2} + \frac{25\lambda_2}{2} + \frac{2\lambda_1}{2} + \frac{4\lambda_2}{2} + \frac{3\lambda_1}{2} + \frac{\lambda_2}{2} = 5$

$$10\lambda_1 + 30\lambda_2 = 10 \longrightarrow (7)$$

for optimization

$$H^B = \begin{pmatrix} 0 & P \\ P^T & Q \end{pmatrix}$$

$$P = \begin{pmatrix} \frac{\partial h_1}{\partial x_1} & \frac{\partial h_1}{\partial x_2} & \frac{\partial h_1}{\partial x_3} \\ \frac{\partial h_2}{\partial x_1} & \frac{\partial h_2}{\partial x_2} & \frac{\partial h_2}{\partial x_3} \end{pmatrix}$$

$$Q = \begin{pmatrix} \frac{\partial^2 v}{\partial x_1^2} & \frac{\partial^2 v}{\partial x_1 \partial x_2} & \frac{\partial^2 v}{\partial x_1 \partial x_3} \\ \vdots & \vdots & \vdots \\ P & & Q \end{pmatrix}_{5\times 5}$$

$$H^B = \begin{pmatrix} 0 & 0 & 0 & & \\ 0 & 0 & 0 & & \\ 0 & 0 & 0 & & \\ & & & Q & \\ & & & & \end{pmatrix}$$

By solving (6) & (7)

$\lambda_1 = 0.08$

$\lambda_2 = 0.3$

(1) $\Rightarrow x_1 = \frac{0.08 + 5(0.3)}{2} = 0.79.$

$x_2 = \frac{0.08 + 2(0.3)}{2} = 0.34.$

$x_3 = \frac{(0.08)3 + 0.3}{2} = 0.27.$

$f(x) = (0.79)^2 + (0.34)^2 + (0.27)^2 = 0.8126$ at $x_1 = 0.79$, $x_2 = 0.34$ $x_3 = 0.27$.

Constrained optimization with inequality constrains.

Consider the general NLPP problem as follows

optimize $Z = f(x_1, x_2, \ldots x_n)$

subject to constrains

$$g(x_1, x_2, x_3 \ldots x_n) \leq b$$

$$x_1, x_2, x_3 \ldots x_n \geq 0.$$

Here $h(x_1, x_2, \ldots x_n) = g(x_1, x_2, \ldots x_n) - b$

The Legrangian function can be constructed as

$$L(x, s, \lambda) = f(x) - \lambda(h(x) + s^2)$$

The necessary conditions for the maximum is restated as

$$\frac{\partial f}{\partial x_i} - \lambda \frac{\partial h}{\partial x_i} = 0$$

$$\lambda h(x) = 0$$

$$h(x) \leq 0$$

$$\lambda \geq 0.$$

These necessary conditions are called Kuhn-Tucker conditions

Sufficient condition for optimization

The Kuhn-Tucker condition for a maximization NLPP subject to contrains $h(x) \leq 0$ and $x \geq 0$ are also sufficent condition for the maximum of $f(x)$. If $f(x)$ is concave and $h(x)$ is convex

1) Maximize $Z = 4x_1 + x_1^3 + 2x_2$

Subject to $x_1 + x_2 \leq 1$

$x_1, x_2 \geq 0$

Soln:

$f(x) = 4x_1 + x_1^3 + 2x_2$

$g(x) = x_1 + x_2, \quad b = 1$

$h(x) = x_1 + x_2 - 1$

The Kuhn-Tucker conditions are

$$\frac{\partial f}{\partial x_1} - \lambda \frac{\partial h}{\partial x_1} = 0$$

$$\frac{\partial f}{\partial x_2} - \lambda \frac{\partial h}{\partial x_2} = 0$$

$\lambda h(x) = 0$

$h(x) \leq 0$

$\lambda \geq 0$

$4 - 3x_1^2 - \lambda = 0$

$4 = 3x_1^2 + \lambda \rightarrow (1)$

$2 - \lambda = 0$

$\boxed{\lambda = 2}$ → (2).

$\lambda(x_1 + x_2 - 1) = 0$

$\lambda = 0$ (or) $x_1 + x_2 - 1 = 0$ → (3)

$x_1 + x_2 - 1 \leq 0$ → (4).

(1) ⇒ $4 - 2 = 3x_1^2$

$x_1^2 = 2/3$

$x_1 = \sqrt{2/3}$ since $x_1 \& x_2 \geq 0$.

$x_1 = \sqrt{2/3}$

(3) ⇒ $\sqrt{2/3} + x_2 = 1$

$x_2 = 0.183$ → IP we should do it like (Appendix)

$f = 4x_1 + x_1^3 + 2x_2$

$\frac{\partial f}{\partial x_1} = 4 - 3x_1^2$

$\frac{\partial^2 f}{\partial x_1^2} = -6x_1$ (A) < 0

$\frac{\partial^2 f}{\partial x_1 x_2} = 0$ (B)

$\frac{\partial f}{\partial x_2} = 2$

$\frac{\partial^2 f}{\partial x_2^2} = 0$ (C)

$AC - B^2 = 0$. $(-6x_1)(0) - 0 = 0$.

f is strictly concave since $\frac{\partial^2 f}{\partial x_1^2} < 0$.

$\frac{\partial h}{\partial x_1} = 1 = \frac{\partial h}{\partial x_2}$

$\frac{\partial^2 h}{\partial x_1^2} = 0 = \frac{\partial^2 h}{\partial x_2^2}$ (A & C)

$\frac{\partial^2 h}{\partial x_1 \partial x_2} = 0$ (B)

$\frac{\partial f}{\partial x} = 4 + 3x_1^2$

$\frac{\partial^2 f}{\partial x_1^2} = 6x_1$

$\frac{\partial f}{\partial x_2} = 2$

$\frac{\partial f}{\partial x_2 \partial x_1} = 0$

$(6x_1)(0) - 0 = 0$

$AC - B^2 = (0) - 0 = 0$.

h is convex

Max $z = 4(\sqrt{2/3}) - (\sqrt{2/3})^3 + 2(0.183)$.

Max $Z = 3.087$ at $x_1 = \sqrt{2/3}$ and $x_2 = 0.183$.

$g(x) \Rightarrow x_1 + x_2 - 1 \leq 0$. IP

$\sqrt{2/3} + 0.183 - 1 \Rightarrow -0.001 \leq 0$]

(2) Maximize $Z = 10x_1 + 4x_2 - 2x_1^2 - x_2^2$
subject to $2x_1 + x_2 \leq 5$ and
$x_1, x_2 \geq 0$.

Soln.

$f(x) = 10x_1 + 4x_2 - 2x_1^2 - x_2^2$
$g(x) = 2x_1 + x_2$, $b = 5$
$h(x) = 2x_1 + x_2 - 5$

The Kuhn-Tucker conditions are

$$\frac{\partial f}{\partial x_1} - \lambda \frac{\partial h}{\partial x_1} = 0$$

$$\frac{\partial f}{\partial x_2} - \lambda \frac{\partial h}{\partial x_2} = 0.$$

$\lambda h(x) = 0$
$h(x) \leq 0$.
$\lambda \geq 0$.
$10 - 4x_1 - \lambda x = 0$.

$10 - 4x_1 = 2\lambda \longrightarrow (1)$

$4 - 2x_2 - \lambda = 0$

$4 - 2x_2 = \lambda \longrightarrow (2)$

$\lambda(2x_1 + x_2 - 5) = 0$

$\lambda = 0$ (or) $2x_1 + x_2 - 5 = 0 \longrightarrow (3)$

$2x_1 + x_2 - 5 \leq 0 \longrightarrow (4)$.

case(i) $\lambda = 0$.

$(1) \Rightarrow 10 - 4x_1 = 0$.

$10 = 4x_1$

$x_1 = 5/2$

$(2) \Rightarrow 4 - 2x_2 = 0$

$4 = 2x_2$

$x_2 = 2$.

$(4) \Rightarrow 2 \times 5/2 + 2 = 7 \leq 0$. is not possible.

case(ii)

$2x_1 + x_2 = 5 \longrightarrow (5)$.

$(1) \Rightarrow 10 - 4x_1 = 2\lambda$

$(2) \times 2 \Rightarrow 8 - 4x_2 = 2\lambda$

$2 - 4x_1 + 4x_2 = 0$.

$4x_1 - 4x_2 = 2$

$2x_1 - 2x_2 = 1 \longrightarrow (6)$.

$(5) - (6)$

$3x_2 = 4$

$x_2 = 4/3$

$(5) \Rightarrow 2x_1 + 4/3 = 5$

$2x_1 = 11/3$

$x_1 = 1.83$.

$(4) \Rightarrow 2(1.83) + 4/3 - 5 \leq 0$.

Feasible.

$\frac{\partial f}{\partial x_1} = 10 - 4x_1$

$\frac{\partial^2 f}{\partial x_1^2} = -4 \quad (A)$

$\frac{\partial^2 f}{\partial x_1 \partial x_2} = 0 \quad (B)$

$\frac{\partial f}{\partial x_2} = 4 - 2x_2$

$\frac{\partial^2 f}{\partial x_2^2} = -2 \quad (C)$

$AC - B^2 = 8 - 0 > 0$.

A and C < 0

f is concave

$\frac{\partial h}{\partial x_1} = 2 \quad , \frac{\partial h}{\partial x_2} = 1$

$\frac{\partial^2 h}{\partial x_1^2} = 0 \quad , \frac{\partial^2 h}{\partial x_2^2} = 0 \quad , \frac{\partial^2 h}{\partial x_1 \partial x_2} = 0$

$AC - B^2 = 0$.

h is convex.

Max $Z = 10(1.83) + 4(4/3) - 2(1.83)^2 - (4/3)^2$

$= 11.161$ at $x_1 = 1.83$ and $x_2 = 4/3$.

15.18

1.333

1.776.

3.348.

Maximize $Z = 8x_1 + 10x_2 - x_1^2 - x_2^2$

subject to

$3x_1 + 2x_2 \leq 6$

$x_1, x_2 \geq 0$

(3) Minimize $z = 0.3x_1^2 - 2x_1 + 0.4x_2^2 - 2.4x_2 + 0.6x_1x_2 + 100$

subject to constrains

$$2x_1 + x_2 \geq 4$$

$$x_1, x_2 \geq 0.$$

Soln:

$$f(x) = 0.3x_1^2 - 2x_1 + 0.4x_2^2 - 2.4x_2 + 0.6x_1x_2 + 100$$

$$g(x) = 2x_1 + x_2, \quad b = 4$$

$$h(x) = 2x_1 + x_2 - 4.$$

The Kuhn-Tucker conditions are

$$\frac{\partial f}{\partial x_1} - \lambda \frac{\partial h}{\partial x_1} = 0$$

$$\frac{\partial f}{\partial x_2} - \lambda \frac{\partial h}{\partial x_2} = 0.$$

$$\lambda h(x) = 0$$

$$h(x) \leq 0$$

$$\lambda \geq 0.$$

$$0.6x_1 - 2 + 0.6x_2 - 2\lambda = 0 \longrightarrow (1)$$

$$0.8x_2 - 2.4 + 0.6x_1 - \lambda = 0 \longrightarrow (2)$$

$$\lambda(2x_1 + x_2 - 4) = 0 \longrightarrow (3)$$

$$2x_1 + x_2 - 4 \leq 0 \longrightarrow (4)$$

From (3) $\lambda = 0$, $2x_1 + x_2 - 4 = 0$

case(i) $\lambda = 0$

$$0.6x_1 + 0.6x_2 = 2$$

$$0.8x_2 + 0.6x_1 = 2.4$$

$$(-) \quad (-) \quad (-)$$

$$-0.2x_2 = -0.4$$

$$\boxed{x_2 = 2}$$

$0.6x_1 + 0.6(0.2) = 2$

$\boxed{x_1 = 1.33}$

④ $\Rightarrow 2(1.33) + 2 - 4 > 0$.

which contradictory to equation ④.

$2x_1 + x_2 = 4 \longrightarrow ⑤$.

① $\Rightarrow 0.6x_1 + 0.6x_2 - 2\lambda = 2$

② $\times 2 \Rightarrow \underset{(-)}{1.2x_1} + \underset{(-)}{1.6x_2} \underset{(+)}{- 2\lambda} = \underset{(-)}{4.8}$

$-0.6x_1 - x_2 = -2.8 \longrightarrow ⑥$.

Solve ⑤ & ⑥

$x_1 = 0.85 \qquad x_2 = 2.28$.

④ $\Rightarrow$

$= 2(0.85) + 2.28 - 4$

$= -0.02 \leq 0$

Feasible.

$\frac{\partial f}{\partial x_1} = 0.6x_1 - 2 + 0.6x_2$

$\frac{\partial^2 f}{\partial x_1^2} = 0.6 \ (A)$

$\frac{\partial^2 f}{\partial x_1 \partial x_2} = 0.6 \ (B)$

$\frac{\partial f}{\partial x_2} = 0.8x_2 - 2.4 + 0.6x_1$

$\frac{\partial^2 f}{\partial x_2^2} = 0.8 \ (C)$

$AC - B^2 = 0.48 - 0.36 > 0$

A and C > 0

f is convex

$\frac{\partial h}{\partial x_1} = 2$, $\frac{\partial^2 h}{\partial x_1^2} = 0$ (A) $\frac{\partial^2 h}{\partial x_1 \partial x_2} = 0$ (B)

$\frac{\partial h}{\partial x_2} = 1$, $\frac{\partial^2 h}{\partial x_2^2} = 0$ (C)

$AC - B^2 = 0$.

A & C is zero

h is concave

$\min z = 0.3(0.85)^2 - 2(0.85) + 0.4(2.28)^2 - 2.4(2.28) + 0.6(0.85)(2.28) +$ (0.85 → 0.7225, 2.28 → 5.1984)

100

$= 96.305$ at $x_1 = 0.85$, $x_2 = 2.28$.

constrained optimization with more than one inequality constraint.

Consider the non-LPP Max $z = f(x_1, x_2 \ldots x_n)$

subject to constraints $g_i(x_1, x_2 \ldots) \leq b_i$

$x_1, x_2, \ldots \geq 0$.

The necessary condition for f to be maximum is

$$\frac{\partial f}{\partial x_1} - \lambda_1 \frac{\partial h_1}{\partial x_1} - \lambda_2 \frac{\partial h_2}{\partial x_1} = 0.$$

where $h_1 = g_1(x_1, \ldots x_n) - b$, $h_2 = g_2(x_1 \ldots x_n) - b$

$$\frac{\partial f}{\partial x_2} - \lambda_1 \frac{\partial h_1}{\partial x_2} - \lambda_2 \frac{\partial h_2}{\partial x_2} = 0$$

$\lambda_1 (h_1) = 0$

$\lambda_2 h_2 = 0$.

$h_1 \leq 0$

$h_2 \leq 0$.

where λ_1 & λ_2 are Lagrangian multipliers.

NOTE:

For maximization problem the constraints is of the type

less
greater than (or) equal to. If the constraints are greater than (or) equal to type then convert to less than (or) equal to (multiple with negative on both sides).

For minimization problem the constraint must be greater than (or) equal to type

Sufficiency of Kuhn - Tucker

For the NLPP of maximizing $f(x_1, \ldots x_n)$, subject to the inequality $h_i(x_1, \ldots x_n) \leq 0$, the Kuhn-Tucker conditions are also sufficient for maximum if $f(x)$ is concave and all h_i's are convex

Solve the following NLPP

Max $Z = 7x_1^2 + 8x_1 + 5x_2^2$

Subject to

$x_1 + 2x_2 \leq 10$

$x_1 - 3x_2 \leq 9$, $x_1, x_2 \geq 0$.

Soln:

$f(x) = 7x_1^2 + 8x_1 + 5x_2^2$

$h_1(x) = x_1 + 2x_2 - 10$

$h_2(x) = x_1 - 3x_2 - 9$

The Kuhn Tucker conditions are

$$\frac{\partial f}{\partial x_1} - \lambda_1 \frac{\partial h_1}{\partial x_1} - \lambda_2 \frac{\partial h_2}{\partial x_2} = 0.$$

$$\frac{\partial f}{\partial x_2} - \lambda_1 \frac{\partial h_1}{\partial x_2} - \lambda_2 \frac{\partial h_2}{\partial x_2} = 0$$

$\lambda_1 (h_1) = 0$

$\lambda_2 (h_2) = 0$

$h_1 \leq 0$

$h_2 \leq 0$.

1.0609 20.0704

$14x_1 + 8 - \lambda_1 - \lambda_2 = 0 \longrightarrow (1)$

$10x_2 - 2\lambda_1 + 3\lambda_2 = 0 \longrightarrow (2)$

$h_1\lambda_1 = 0, \quad h_2\lambda_2 = 0$

$\lambda_1(x_1 + 2x_2 - 10) = 10 \longrightarrow (3)$

$\lambda_2(x_1 + 3x_2 - 9) = 0 \longrightarrow (4)$

$x_1 + 2x_2 - 10 \leq 0 \longrightarrow$

$x_1 - 3x_2 - 9 \leq 0$

Case (i)

$\lambda_1 = \lambda_2 = 0$

$(1) \Rightarrow 14x_1 + 8 = 0$

$(2) \Rightarrow 10x_2 = 0$

$14x_1 = -8$

$x_1 = -8/14$

$x_2 = 0.$

Not feasible

Case (iii)

$\lambda_1 \neq 0, \quad \lambda_2 = 0.$

$(1) \Rightarrow 14x_1 + 8 - \lambda_1 = 0$

$(2) \Rightarrow 28x_1 - 2\lambda_1 = -16$

$10x_2 - 2\lambda_1 = 0$

$28x_1 - 10x_2 = -16$

$(5) \Rightarrow$

$x_1 + 2x_2 = 10.$

Solving

$x_1 = 1.03$

$x_2 = 4.48. \quad \lambda_1 = 22.42$

It is feasible

Case (ii)

$\lambda_1 = 0, \quad \lambda_2 \neq 0.$

$(1) \Rightarrow 14x_1 + 8 - \lambda_2 = 0 \longrightarrow (5)$

$(2) \Rightarrow 10x_2 + 3\lambda_2 = 0 \longrightarrow (6)$

$(4) \Rightarrow x_1 - 3x_2 = 9 \longrightarrow (7)$

$(5) \times 3$

$42x_1 - 3\lambda_2 = -24$

$10x_2 + 3\lambda_2 = 0$

$42x_1 + 10x_2 = -24 \longrightarrow (8)$

Solve (7) & (8)

$x_1 = 0.13$

$x_2 = -2.9 \qquad \lambda_2 = 9.82$

Not feasible.

$\max Z = 7(1.03)^2 + 8(1.03) + 5(4.48)^2$

$\max Z = 116.01$

Case(iv)

$\lambda_1 \neq \lambda_2 \neq 0$.

$14x_1 + 8 - \lambda_1 - \lambda_2 = 0$

$10x_2 - 2\lambda_1 + 3\lambda_2 = 0$

$x_1 + 2x_2 = 10$

$x_1 - 3x_2 = 9$

$x_1 = 9.6$, $x_2 = 0.21$

$14(9.6) - \lambda_1 - \lambda_2 + 8 = 0$.

$\lambda_1 + \lambda_2 = 142.4 \longrightarrow (9)$.

$10(0.21) - 2\lambda_1 + 3\lambda_2 = 0$.

$2\lambda_1 - 3\lambda_2 = 2.1 \longrightarrow (10)$.

Solve (9) & (10)

$\lambda_1 = 85.6$

$\lambda_2 = 56.5$

It is feasible.

Max $Z = 7(9.6)^2 + 8(9.6) + 5(0.21)^2$.

$= 7(92.16) + 8(9.6) + 5(0.0441)$

$= 645.12 + 76.8 + 0.2205$.

$= 722.4$.

Minimize $Z = 2x_1 + 3x_2 - x_1^2 - 2x_2^2$

Sub to constraints

$$x_1 + 3x_2 \leq 6$$

$$5x_1 + 2x_2 \leq 10 \quad , x_1, x_2 \geq 0$$

Soln :-

$$f(x) = 2x_1 + 3x_2 - x_1^2 - 2x_2^2$$

$$h_1(x) = x_1 + 3x_2 - 6$$

$$h_2(x) = 5x_1 + 2x_2 - 10$$

The Kuhn-Tucker conditions are

$$\frac{\partial f}{\partial x_1} - \lambda_1 \frac{\partial h_1}{\partial x_1} - \lambda_2 \frac{\partial h_2}{\partial x_1} = 0$$

$$\frac{\partial f}{\partial x_2} - \lambda_1 \frac{\partial h_1}{\partial x_2} - \lambda_2 \frac{\partial h_2}{\partial x_2} = 0$$

$$\lambda_1 h_1 = 0$$

$$\lambda_2 h_2 = 0$$

$$h_1 \leq 0$$

$$h_2 \leq 0$$

$$2 - 2x_1 - \lambda_1 - 5\lambda_2 = 0 \longrightarrow (1)$$

Optimize (or) solve the following NLPP

$$Z = 2x_1 + 3x_2 - (x_1^2 + x_2^2 + x_3^2)$$

Subject to constraint

$$x_1 + x_2 \leq 1$$

$$2x_1 + 3x_2 \leq 6$$

$$x_1, x_2 \geq 0.$$

Soln.

$$f(x) = 2x_1 + 3x_2 - (x_1^2 + x_2^2 + x_3^2)$$

$$h_1(x) = x_1 + x_2 - 1 = 0$$

$$h_2(x) = 2x_1 + 3x_2 - 6 = 0$$

To find optimization we construct the Hessian matrix as follows.

$$H^B = \begin{pmatrix} O & P \\ P^T & Q \end{pmatrix}_{(m+n)\times(m+n)}$$

where $m \rightarrow$ no. of constraints $= 2$.

$n \rightarrow$ no. of variables $= 3$

O — $m \times m$ null matrix

$$O = \begin{pmatrix} 0 & 0 \\ 0 & 0 \end{pmatrix}_{2\times 2}$$

$$P = \begin{pmatrix} \frac{\partial h_1}{\partial x_1} & \frac{\partial h_1}{\partial x_2} & \frac{\partial h_1}{\partial x_3} \\ \frac{\partial h_2}{\partial x_1} & \frac{\partial h_2}{\partial x_2} & \frac{\partial h_2}{\partial x_3} \end{pmatrix} = \begin{pmatrix} 1 & 1 & 0 \\ 2 & 3 & 0 \end{pmatrix}$$

$$Q = \begin{pmatrix} \frac{\partial^2 L}{\partial x_1^2} & \frac{\partial^2 L}{\partial x_1 \partial x_2} & \frac{\partial^2 L}{\partial x_1 \partial x_3} \\ \frac{\partial^2 L}{\partial x_2 \partial x_1} & \frac{\partial^2 L}{\partial x_2^2} & \frac{\partial^2 L}{\partial x_2 \partial x_3} \\ \frac{\partial^2 L}{\partial x_3 \partial x_1} & \frac{\partial^2 L}{\partial x_2 \partial x_2} & \frac{\partial^2 L}{\partial x_3^2} \end{pmatrix}$$

$$L = f(x) - \lambda_1 h_1(x) - \lambda_2 h_2(x)$$
$$= 2x_1 + 3x_2 - (x_1^2 + x_2^2 + x_3^2) - \lambda_1 (x_1 + x_2 - 1) - \lambda_2 (2x_1 + 3x_2 - 6)$$

$$\frac{\partial L}{\partial x_1} = 2 - 2x_1 - \lambda_1 - 2\lambda_2$$

$$\frac{\partial^2 L}{\partial x_1^2} = -2$$

$$\frac{\partial L}{\partial x_2} = 3 - 2x_2 - \lambda_1 - 3\lambda_2$$

$$\frac{\partial^2 L}{\partial x_2^2} = -2$$

$$\frac{\partial L}{\partial x_3} = -2x_3$$

$$\frac{\partial^2 L}{\partial x_3^2} = -2$$

$$\frac{\partial^2 L}{\partial x_1 \partial x_2} = \frac{\partial^2 L}{\partial x_1 \partial x_3} = \frac{\partial^2 L}{\partial x_2 \partial x_1} = \frac{\partial^2 L}{\partial x_2 \partial x_3} = \frac{\partial^2 L}{\partial x_3 \partial x_1} = \frac{\partial^2 L}{\partial x_3 \partial x_2} = 0.$$

$$Q = \begin{pmatrix} -2 & 0 & 0 \\ 0 & -2 & 0 \\ 0 & 0 & -2 \end{pmatrix}$$

$$H^B = \begin{pmatrix} 0 & 0 & 1 & 1 & 0 \\ 0 & 0 & 2 & 3 & 0 \\ 1 & 2 & -2 & 0 & 0 \\ 1 & 3 & 0 & -2 & 0 \\ 0 & 0 & 0 & 0 & -2 \end{pmatrix}$$

$$|H^B| = 1 \begin{vmatrix} 0 & 0 & 3 & 0 \\ 1 & 2 & 0 & 0 \\ 1 & 3 & -2 & 0 \\ 0 & 0 & 0 & -2 \end{vmatrix} - 1 \begin{vmatrix} 0 & 0 & 2 & 0 \\ 1 & 2 & -2 & 0 \\ 1 & 3 & 0 & 0 \\ 0 & 0 & 0 & -2 \end{vmatrix} = -2.$$

Sign of $H^B = (-1)^{m+n} = (-1)^5 = -ve \Rightarrow max$

sign of $H^B = (-1)^m = (-1)^2 = +ve \Rightarrow min$

Graphical Method

Determine the value of x_1 & x_2 so as to minimize $Z = x_1^2 + x_2^2$

subject to $x_1 + x_2 \geq 8$

$x_1 + 2x_2 \geq 10$

$2x_1 + x_2 \geq 10$

$x_1, x_2 \geq 0$

Soln:

I $x_1 + x_2 = 8$

$x_1 = 0, \; x_2 = 8$

$x_2 = 0, \; x_1 = 8$

$(0, 8), (8, 0)$

ii $x_1 + 2x_2 = 10$

$x_1 = 0, \; x_2 = 5$

$x_2 = 0 \quad x_1 = 10$

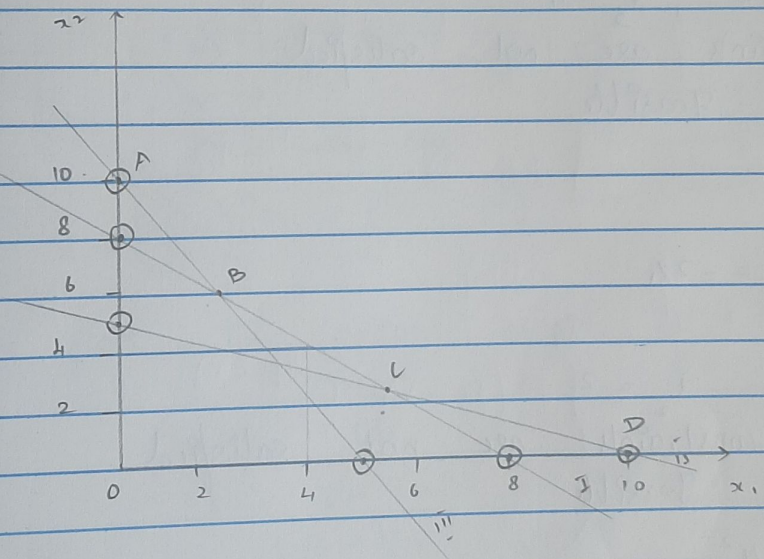
$(0, 5) \; (10, 0)$

iii $2x_1 + x_2 = 10$

$x_1 = 0 \quad x_2 = 10$

$x_2 = 0 \quad x_1 = 5$

$(0, 10) \; (5, 0)$



$x_1^2 + x_2^2 = r^2$

$2x_1 dx_1 + 2x_2 dx_2 = 0$

$x_1 dx_1 = -x_2 dx_2$

$\frac{dx_2}{dx_1} = -x_1/x_2 \longrightarrow (1)$

I $dx_1 + dx_2 = 0$

$dx_1 = -dx_2$

$\frac{dx_2}{dx_1} = -1 \longrightarrow (2)$

ii $dx_1 + 2dx_2 = 0$

$dx_1 = -2dx_2$

$dx_2/dx_1 = -1/2 \longrightarrow (3)$

iii) $2dx_1 + dx_2 = 0$

$$\frac{dx_2}{dx_1} = -2/1 \longrightarrow (4)$$

(1) & (2)

$$\frac{-x_1}{x_2} = -1 \qquad x_1 = x_2 \Rightarrow x_1 = x_2 = 4$$

All the contradinst are satisfied

Feasible.

(1) & (3)

$$\frac{-x_1}{x_2} = -1/2$$

$x_1 = 2$, $x_2 = 4$

The constraints are not satisfied

Not Feasible

(1) & (4)

$$\frac{-x_1}{x_2} = -2/1$$

$x_1 = 4 \quad x_2 = 2$

All the constraints are not satisfied

Not Feasible

min $Z = 4^2 + 4^2$

$= 32$ at $x_1 = x_2 = 4$.

Unit-5

Queuing Theory

2m Kendal's notation for ~~xepresde~~ representing Queuing Theory.

Queing model may be completely specified in the following symbol form

$$(a/b/c):(d/e)$$

where a - probability law for arrival, b - probability law according to which customers are served, c - no. of channels (or) service station, d - capacity of the system, e - Queue discipline

Model-1 : $(M/M/\underset{\text{Roman letter}}{I}):(\infty/FIFO)$

$P_0 = 1 - \frac{\lambda}{\mu}$ ($\mu \rightarrow$ average time)

$$P = \frac{\lambda}{\mu} < 1$$

$$P_n \text{ (or) } P^n = \left(\frac{\lambda}{\mu}\right)^n P_0 = P^n(1-P)$$

1 => To find the average number of units in the system

$$L_s = \frac{\lambda/\mu}{1-\lambda/\mu}$$

$$L_s = \frac{\lambda/\mu}{(\mu-\lambda)/\mu} = \frac{\lambda}{\mu-\lambda} \text{ (or) } \left(\frac{P}{1-P}\right)$$

2 => To find the average length of Queue

$$L_q = L_s - \frac{\lambda}{\mu}.$$

$$L_q = \frac{P^2}{1-P}$$

3 => Expected waiting time in the system

$$W_s = \frac{1}{\mu-\lambda}$$

$$P(w>0) = 1 - (1 - \lambda/\mu) = 1 - \left(\frac{\mu - \lambda}{\mu}\right)$$

$$P(w>0) = \lambda/\mu = \rho \qquad \left(\frac{\mu - \mu + \lambda}{\mu}\right)$$

4⇒ Waiting line in the Queue

$$w_q = \frac{\lambda}{\mu(\mu - \lambda)}$$

5⇒ Expected waiting time of a customer who has to wait

$$(w/w>0) = \frac{1}{\mu - \lambda}$$

6⇒ Expected length of the non-empty Queue

$$(L/L>0) = \frac{\mu}{\mu - \lambda}$$

7⇒ Probability of Queue size greater than N

$$P_{\ldots>N} = \rho^N$$

8⇒ Probability of waiting time in the system greater than equal to t

$$= \int_t^\infty (\mu - \lambda)\, e^{-(\mu-\lambda)w}\, dw$$

9⇒ Probability of waiting time in the queue greater than equal to t

$$= \int_t^\infty \rho(\mu - \lambda)\, e^{-(\mu-\lambda)w}\, dw$$

10⇒ Traffic intensity

$$\frac{\lambda}{\mu}$$

1) In a railway marshalling yard, goods train arrive at the rate of 30 trains per day. Assuming that inter arrival time follows an exponential distribution and the service time distribution is also exponential, with an average of 36 minutes. Calculate the following

i) The mean Queue size l_q, l_s.

ii) The probability that Queue size exceeds 10 P^{10}

iii) If the input of the train increases to an average 33 per day what will be changes in (i) & (ii)

Soln:

Arrival rate $\lambda = \frac{30}{24 \times 60} = \frac{1}{48}$ per minute

$\mu = 36$ minutes $\Rightarrow \frac{1}{36}$ per minute.

i) Mean Queue size (L_q)

$$L_s = \frac{\lambda}{\mu - \lambda} = \frac{1/48}{1/36 - 1/48} = 3$$

0.027 -

$$L_q = L_s - \frac{\lambda}{\mu} = 2.25 \approx 3.$$

ii) The probability that Queue size exceeds 10

$$(P \geq 10) = \rho^{10}$$

$$= \left(\frac{\lambda}{\mu}\right)^{10}$$

$$= (0.75)^{10}$$

$$(P \geq 10) = 0.056$$

iii) The input increases to 33 trains per day.

$$\lambda = \frac{33}{24 \times 60} = 0.0229 \text{ per minute}$$

$$\mu = \frac{1}{36} \text{ per minute.}$$

i) $$Ls = \frac{\lambda}{\mu - \lambda} = \frac{0.0229}{0.027 - 0.022}$$

$$= 4.6$$

$$\simeq 5$$

ii) $$(P \geq 10) = \rho^{10}$$

$$= \left(\frac{\lambda}{\mu}\right)^{10}$$

$$= \left(\frac{0.0229}{0.027}\right)^{10}$$

$$= 0.146$$

In a supermarket the average arrival rate of customers is 10 every 30 minutes following poission process. The average time taken by the cashier to list and calculate the customer purchase is 2.5 minutes following exponential distribution
i) What is the probability that the queue length exceeds 6? ρ^6
ii) What is the expected time by the customer in the system? Ws

Soln.

arrival time $$\lambda = \frac{10}{30} \text{ per minute}$$

$$\mu = \frac{1}{2.5} \text{ per minute}$$

$$\rho = (0.8333)$$

Probability of the Queue size exceeds 6

i) $(P > 6) = \rho^6$

$= (\lambda/\mu)^6$

$= (0.8333)^6$

$= 0.3348$

ii)

$$W_s = \frac{1}{\mu - \lambda}$$

$$= \frac{1}{12.5 - \frac{10}{30}}$$

$W_s = 15$ minutes

In a public telephone booth the arrivals are on the average 15 per hour. A call on the average takes 3 minutes. If there is just one phone find

i) Expected no. of callers in the booth at any time $(L/L>0)$

ii) The proportion of the time the booth is expected to be idle $\rho, 1-\rho$

Soln:

$\lambda = 15$ per hour

$\mu = \frac{1}{3} \times 60 = 20$ per hour

$\rho = \lambda/\mu = \frac{15}{20}$

i) The Expected Length of the non-Empty Queue

$$(L/L>0) = \frac{\mu}{\mu - \lambda} = \frac{20}{20 - 15}$$

$= 4.$

ii) The service is busy then

$$\rho = \lambda/\mu = \frac{15}{20} = 0.75$$

The booth expected to ideal

$$1 - 0.75 = \frac{1}{4} \text{ hours}$$

model 1

Customers arrive at a one window drive in bank according to poission distribution with mean 10 per hour. Service time per customer is exponential with mean 5 minutes. The space infront of the window including that for the serviced car can accomodate a maximum of 3 cars. Other can wait outside of this space.

$P_0 + P_1 + P_2$

i) What is the probability that an arriving customer can drive directly to the space infront of the window.

ii) What is the probability that an arriving customer will have to wait outside the indicated the space. $1 - [P_0 + P_1]$

iii) How long is arriving customer is expected to wait before starting service

Soln:

$\lambda = 10$ per hour

$\mu = \frac{1}{5}$ per minute $\Rightarrow \frac{1}{5} \times 60 \Rightarrow 12$ per hour.

i) The probability that an arriving customer can drive directly to the space infront of the window.

$P_0 + P_1 + P_2$

$P_n = \rho^n P_0$

$= \rho^n (1 - \rho)$

$\rho = \frac{\lambda}{\mu} = \frac{10}{12} = 0.8333$

$$P_0 + P_1 + P_2 = (1-\rho) + \rho(1-\rho) + \rho^2(1-\rho)$$

$$= (1-\rho)[1+\rho+\rho^2]$$

$$= (1-0.8333) * [1 + 0.8333 + (0.8333)^2]$$

$$= 0.4212$$

ii) Probability that the arriving customer will have to wait

$$1 - [P_0 + P_1 + P_2] \Rightarrow 1 - 0.4212$$

$$= 0.57$$

iii) Arriving customer is expected to wait before starting service

$$W_q = \frac{\lambda}{\mu(\mu-\lambda)}$$

$$= \frac{10}{12(12-10)}$$

$$W_q = 0.416$$

Car arrives at a petrol pump having one petrol unit in poission fasion with an average of 10 cars per hour. The service time is distributed exponentially with the mean of 3 minutes

i) Average no. of cars in the System. L_S

ii) Average waiting time in the Queue W_q

iii) Average queue length L_q.

iv) The probability that the no. of cars in the system is 2 P^n

Soln:

$\lambda = 10$ per hour

$\mu = \frac{1}{3}$ per minute $= \frac{60}{3}$ per hour $= 20$ per hour.

$$\rho = \frac{\lambda}{\mu} = \frac{10}{20} = 0.5$$

i) Average no. of cars in the system

$$L_s = \frac{\rho}{1-\rho} = \frac{0.5}{1-0.5} = 1 \text{ car}$$

ii) Average waiting time in the queue

$$W_q = \frac{\lambda}{\mu(\mu-\lambda)} = \frac{10}{20(20-10)} = 0.05 \text{ hour}$$

iii) Average queue length

$$L_q = \frac{\rho^2}{1-\rho} = \frac{(0.5)^2}{1-0.5} = 0.5 \text{ car} \simeq 1 \text{ car}$$

iv) Probability that the no. of cars in the system is 2

$$P^n = \rho^n(1-\rho)$$

$n=2$ $= (0.5)^2(1-0.5)$

$= 0.125$

People arrive at the theater tick booth in poission distributed, at the arrival rate of 25 per hour. The service time is constant at 2 minutes. Calculate

i) The mean no. of in the waiting line L_q

ii) The mean waiting time W_q

iii) The ultization factor ρ

Soln

$$\lambda = 25 \text{ per hour}$$

$$\mu = \frac{1}{2} \text{ per minute} = \frac{60}{2} = 30 \text{ per hour}$$

$$\rho = \frac{\lambda}{\mu} = \frac{25}{30} = 0.8333$$

i) The mean no. in the waiting line

$$L_q = \frac{\rho^2}{1-\rho}$$

$$= \frac{(0.8333)^2}{1-0.8333}$$

$$= 4.16$$

$$\approx 4$$

ii) The mean waiting time

$$W_q = \frac{\lambda}{\mu(\mu-\lambda)}$$

$$= \frac{25}{30(30-25)}$$

$$= 0.166$$

iii) The utilization factor

$$\rho = \lambda/\mu = \frac{25}{30} = 0.8333$$

Arrivals at a telephone booth are considered to be possion distribution with an average time of 10 min between one arrival and next, the duration of the phone call is exponential distribution that a person with mean 3 minutes.

i) What is the probability that a person arriving at the booth will have to wait

ii) Find average no. of unit in the system.

iii) Expected waiting time of a customer who has to wait

Soln.

$$\lambda = \frac{1}{10} \text{ per minute}$$

$$\mu = \frac{1}{3} \text{ per minute}$$

$$\rho = \frac{\lambda}{\mu} = \frac{3}{10} = 0.3.$$

iii) Probability that a person arriving at the booth will have to wait

$$(W/W>0) = \frac{1}{\mu - \lambda}$$

$$= \frac{1}{\left(\frac{1}{3} - \frac{1}{10}\right)} = 4.28$$

i)